```
.model NMOS NMOS
+Level = 49

+Lint = 4.4e-08 Tox = 4.4e-09
+Vth0 = 0.3999 Rdsw = 250

+lmin=1.8e-7 lmax=1.8e-7 wmin=1.8e-7 wmax=1.0e-4 Tref=27.0 version =3.1
+Xj= 6.0000000E-08          Nch= 5.9500000E+17
+lln= 1.0000000             lwn= 1.0000000          win= 0.00
+wwn= 0.00                  ll= 0.00
+lw= 0.00                   lwl= 0.00               wint= 0.00
+wl= 0.00                   ww= 0.00                wwl= 0.00
+Mobmod= 1                  binunit= 2              xl=  0
+xw=  0                     binflag=  0
+Dwg= 0.00                  Dwb= 0.00

+K1= 0.5613000              K2= 1.0000000E-02
+K3= 0.00                   Dvt0= 8.0000000         Dvt1= 0.7500000
+Dvt2= 8.0000000E-03        Dvt0w= 0.00             Dvt1w= 0.00
+Dvt2w= 0.00                Nlx= 1.6500000E-07      W0= 0.00
+K3b= 0.00                  Ngate= 5.0000000E+20

+Vsat= 1.3800000E+05        Ua= -7.0000000E-10      Ub= 3.5000000E-18
+Uc= -5.2500000E-11         Prwb= 0.00
+Prwg= 0.00                 Wr= 1.0000000           U0= 3.5000000E-02
+A0= 1.1000000              Keta= 4.0000000E-02     A1= 0.00
+A2= 1.0000000              Ags= -1.0000000E-02     B0= 0.00
+B1= 0.00

+Voff= -0.12350000          NFactor= 0.9000000      Cit= 0.00
+Cdsc= 0.00                 Cdscb= 0.00             Cdscd= 0.00
+Eta0= 0.2200000            Etab= 0.00              Dsub= 0.8000000

+Pclm= 5.0000000E-02        Pdiblc1= 1.2000000E-02  Pdiblc2= 7.5000000E-03
+Pdiblcb= -1.3500000E-02    Drout= 1.7999999E-02    Pscbe1= 8.6600000E+08
+Pscbe2= 1.0000000E-20      Pvag= -0.2800000        Delta= 1.0000000E-02
+Alpha0= 0.00               Beta0= 30.0000000

+kt1= -0.3700000            kt2= -4.0000000E-02     At= 5.5000000E+04
+Ute= -1.4800000            Ua1= 9.5829000E-10      Ub1= -3.3473000E-19
+Uc1= 0.00                  Kt1l= 4.0000000E-09     Prt= 0.00

+Cj= 0.00365                Mj= 0.54                Pb= 0.982
+Cjsw= 7.9E-10              Mjsw= 0.31              Php= 0.841
+Cta= 0                     Ctp= 0                  Pta= 0
+Ptp= 0                     JS=1.50E-08             JSW=2.50E-13
+N=1.0                      Xti=3.0                 Cgdo=2.786E-10
+Cgso=2.786E-10             Cgbo=0.0E+00            Capmod= 2
+NQSMOD= 0                  Elm= 5                  Xpart= 1
+Cgsl= 1.6E-10              Cgdl= 1.6E-10           Ckappa= 2.886
+Cf= 1.069e-10              Clc= 0.0000001          Cle= 0.6
+Dlc= 4E-08                 Dwc= 0                  Vfbcv= -1

.model PMOS PMOS
+Level = 49

+Lint = 3.e-08 Tox = 4.2e-09
+Vth0 = -0.42 Rdsw = 450

+lmin=1.8e-7 lmax=1.8e-7 wmin=1.8e-7 wmax=1.0e-4 Tref=27.0 version =3.1
+Xj= 7.0000000E-08          Nch= 5.9200000E+17
+lln= 1.0000000             lwn= 1.0000000          win= 0.00
+wwn= 0.00                  ll= 0.00
+lw= 0.00                   lwl= 0.00               wint= 0.00
+wl= 0.00                   ww= 0.00                wwl= 0.00
+Mobmod= 1                  binunit= 2              xl= 0.00
+xw= 0.00                   binflag=  0
+Dwg= 0.00                  Dwb= 0.00

+ACM= 0                     ldif=0.00               hdif=0.00
+rsh= 0                     rd= 0                   rs= 0
+rsc= 0                     rdc= 0

+K1= 0.5560000              K2= 0.00
+K3= 0.00                   Dvt0= 11.2000000        Dvt1= 0.7200000
+Dvt2= -1.0000000E-02       Dvt0w= 0.00             Dvt1w= 0.00
+Dvt2w= 0.00                Nlx= 9.5000000E-08      W0= 0.00
+K3b= 0.00                  Ngate= 5.0000000E+20

+Vsat= 1.0500000E+05        Ua= -1.2000000E-10      Ub= 1.0000000E-18
+Uc= -2.9999999E-11         Prwb= 0.00
+Prwg= 0.00                 Wr= 1.0000000           U0= 8.0000000E-03
+A0= 2.1199999              Keta= 2.9999999E-02     A1= 0.00
+A2= 0.4000000              Ags= -0.1000000         B0= 0.00
+B1= 0.00

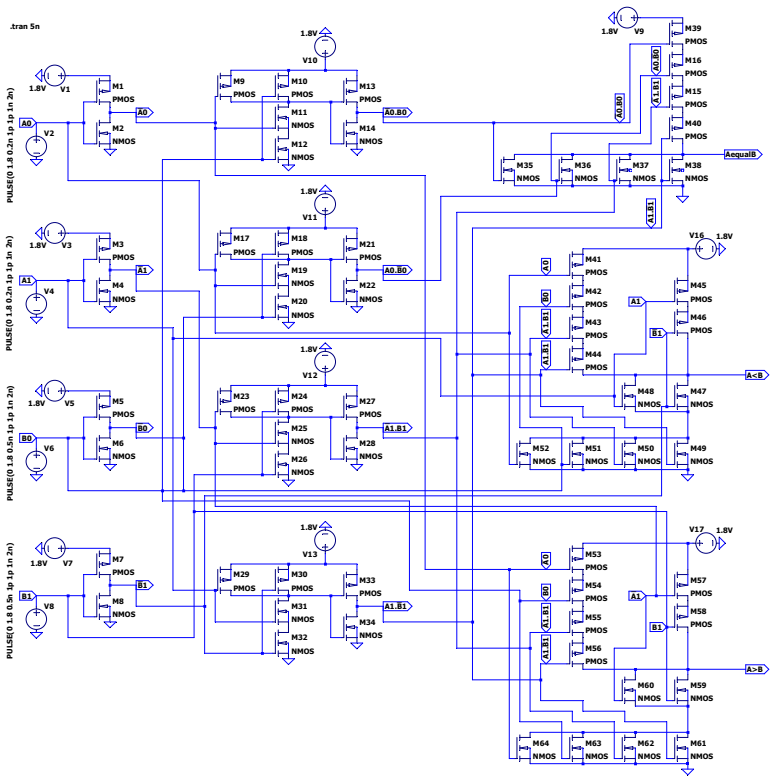
+Voff= -6.40000000E-02      NFactor= 1.4000000      Cit= 0.00
+Cdsc= 0.00                 Cdscb= 0.00             Cdscd= 0.00
+Eta0= 8.5000000            Etab= 0.00              Dsub= 2.8000000

+Pclm= 2.0000000            Pdiblc1= 0.1200000      Pdiblc2= 8.0000000E-05
+Pdiblcb= 0.1450000         Drout= 5.0000000E-02    Pscbe1= 1.0000000E-20
+Pscbe2= 1.0000000E-20      Pvag= -6.0000000E-02    Delta= 1.0000000E-02
+Alpha0= 0.00               Beta0= 30.0000000

+kt1= -0.3700000            kt2= -4.0000000E-02     At= 5.5000000E+04
+Ute= -1.4800000            Ua1= 9.5829000E-10      Ub1= -3.3473000E-19
+Uc1= 0.00                  Kt1l= 4.0000000E-09     Prt= 0.00

+Cj= 0.00138                Mj= 1.05                Pb= 1.24
+Cjsw= 1.44E-09             Mjsw= 0.43              Php= 0.841
+Cta= 0.00093               Ctp= 0                  Pta= 0.00153
+Ptp= 0                     JS=1.50E-08             JSW=2.50E-13
+N=1.0                      Xti=3.0                 Cgdo=2.786E-10
+Cgso=2.786E-10             Cgbo=0.0E+00            Capmod= 2
+NQSMOD= 0                  Elm= 5                  Xpart= 1
+Cgsl= 1.6E-10              Cgdl= 1.6E-10           Ckappa= 2.886
+Cf= 1.058e-10              Clc= 0.0000001          Cle= 0.6
+Dlc= 3E-08                 Dwc= 0                  Vfbcv= -1

.OPTIONS ALLOW_AMBIGUOUS_MODELS
```



yellow